EXPERIMENT-02

AIM: To implement linear search in Python and analyze its best, average, and worst cases using medical records from the Breast Cancer Diagnosis dataset..

Theory:

Linear search is used on a collections of items. It relies on the technique of traversing a list from start to end by exploring properties of all the elements that are found on the way.

For example, consider an array of integers of size . You should find and print the position of all the elements with value . Here, the linear search is based on the idea of matching each element from the beginning of the list to the end of the list with the integer , and then printing the position of the element if the condition is `True'.

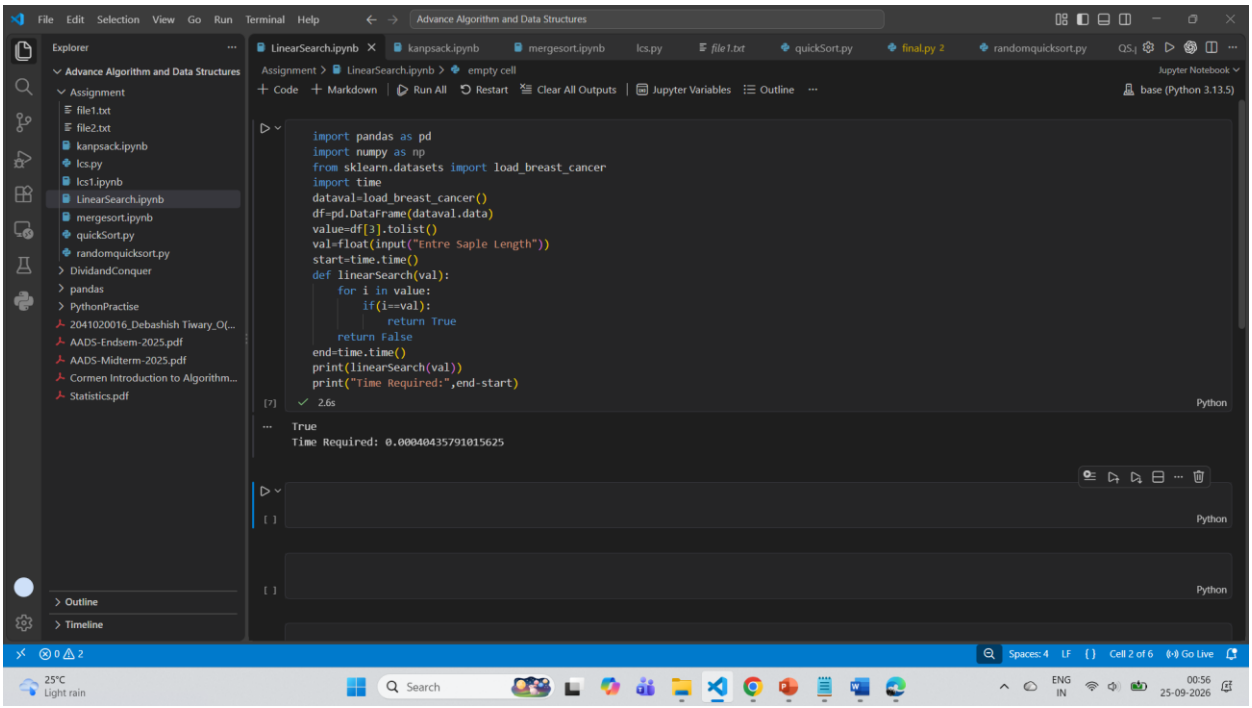
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Code:

```
import pandas as pd
import numpy as np
from sklearn.datasets import load_breast_cancer
import time
dataval=load_breast_cancer()
df=pd.DataFrame(dataval.data)
value=df[3].tolist()
val=float(input("Entre Saple Length"))
start=time.time()
def linearSearch(val):
    for i in value:
        if(i==val):
            return True
    return False
end=time.time()
print(linearSearch(val))
print("Time Required:",end-start)
```

Complexity Analysis:
Best Case = O(1) – If data is present at very first index
Worst Case= O(n) – If data is present at last index or not present at all.

OUTPUT:


File Edit Selection View Go Run Terminal Help
Advance Algorithm and Data Structures
Explorer
Advance Algorithm and Data Structures
Assignment
file1.txt
file2.txt
kanpsack.ipynb
lcs.py
lcs1.ipynb
LinearSearch.ipynb
mergesort.ipynb
quickSort.py
randomquicksort.py
DividandConquer
pandas
PythonPractise
2041020016_Debashish Tiwary_O(...
AADS-Endsem-2025.pdf
AADS-Midterm-2025.pdf
Cormen Introduction to Algorithm...
Statistics.pdf
Outline
Timeline
LinearSearch.ipynb
kanpsack.ipynb
mergesort.ipynb
lcs.py
file1.txt
quickSort.py
final.py 2
randomquicksort.py
Assignment > LinearSearch.ipynb > empty cell
Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
Jupyter Notebook
base (Python 3.13.5)
import pandas as pd
import numpy as np
from sklearn.datasets import load_breast_cancer
import time
dataval=load_breast_cancer()
df=pd.DataFrame(dataval.data)
value=df[3].tolist()
val=float(input("Entre Saple Length"))
start=time.time()
def linearSearch(val):
for i in value:
if(i==val):
return True
return False
end=time.time()
print(linearSearch(val))
print("Time Required:",end-start)
2.6s
Python
True
Time Required: 0.0004043579101625
Python
Python
Spaces: 4 LF { } Cell 2 of 6 Go Live
25°C
Light rain
Search
ENG
IN
00:56
25-09-2026